Department of Artificial Intelligence and Data Science (ADS)

Green University of Bangladesh

Semester: Spring 2026, B.Sc. in ADS

" Functions and Modular Programming "

Lab Report No: 04

Course Title: Introduction to Computational Thinking and Programming

Course Code: ADS 100

Section: 261 D2

Students Details

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Lab Report Status	
Marks:	**Signature:**
Comments:	**Date:**

Lab Experiment Date: 28/03/ 26

Lab Submission Date: 05 /04/ 26

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1. TITLE OF THE LAB REPORT EXPERIMENT

• Write a function named helloWorld() that prints "Hello, World!" when called. Call the function in your main program.

• Create a function called multiplyNumbers(a,b) that takes two numbers as parameters, multiplies them, and returns the result. Call the function and store the result in a variable, then print it.

• Write a modular program that calculates the volume and surface area of a cylinder. Create two separate functions: one for calculating the volume and another for calculating the surface area. In the main program, take user input for the radius and height, then output the results.

• Write a python program to find the largest number between two numbers using function.

• Write a python program to find the sum of the numbers passed as parameters.

• Write a function named checkEvenOdd(n) that checks if a number is even or odd. Inside the function, use an if-else control structure to make the decision. After defining the function, call it in the main program with a sample input and print the result.

• Write functions for performing summation, multiplication, subtraction and division of two numbers.

2. Objective:

The objective of this lab is to develop fundamental programming skills in Python by understanding and applying functions, modular programming, conditional statements, arithmetic operations, and user input/output. The exercises focus on creating reusable functions and organizing programs into logical components to solve different computational problems efficiently.

Specifically, the aims are:

1. To write a function named helloWorld() that prints the message “Hello, World!” and demonstrate how a function is defined and called in a Python program.

2. To create a function multiplyNumbers(a, b) that takes two numbers as parameters, multiplies them, and returns the result, illustrating the concept of parameter passing and returning values from functions.

3. To design a modular program that calculates the volume and surface area of a cylinder by creating two separate functions, demonstrating the concept of dividing a program into smaller reusable modules.

4. To develop a Python program that finds the largest number between two numbers using a function, reinforcing the use of comparison operators and decision-making structures.

5. To write a function that calculates the sum of numbers passed as parameters, showing how functions can process multiple inputs and return computed results.

6. To implement a function checkEvenOdd(n) that determines whether a number is even or odd using an if–else control structure, illustrating the use of conditional logic inside functions.

7. To create separate functions for performing basic arithmetic operations such as summation, subtraction, multiplication, and division of two numbers, demonstrating the concept of reusable functions in solving mathematical problems.

3. PROCEDURE:

The algorithm of the 7 problems is given below.

Problem 01: Print “Hello, World!” using a function

Step 1: Define a function named helloWorld().

Step 2: Inside the function, write a print statement to display "Hello, World!".

Step 3: End the function definition.

Step 4: In the main program, call the function helloWorld().

Step 5: The function executes and prints the message on the screen.

Problem 02: Multiply two numbers using a function

Step 1: Define a function named multiplyNumbers(a, b).

Step 2: Pass two numbers as parameters to the function.

Step 3: Inside the function, multiply the two numbers.

Step 4: Return the multiplication result.

Step 5: In the main program, call the function with two numbers.

Step 6: Store the returned value in a variable.

Step 7: Print the result with a proper message.

Problem 03: Calculate volume and surface area of a cylinder using functions

Step 1: Define a function calculate Volume (radius, height).

Step 2: Use the formula Volume = $\pi \times r^2 \times h$ to calculate the volume.

Step 3: Return the volume.

Step 4: Define another function Surface_Area(radius, height).

Step 5: Use the formula Surface Area = $2 \times \pi \times r \times (r + h)$ to calculate the surface area.

Step 6: Return the calculated surface area.

Step 7: In the main program, take the radius and height as input from the user.

Step 8: Call both functions with the input values.

Step 9: Print the volume and surface area.

Problem 04: Find the largest number between two numbers using a function

Step 1: Define a function named find Largest(a, b).
Step 2: Pass two numbers as parameters to the function.
Step 3: Compare the two numbers using comparison operators.
Step 4: If a > b, return a as the largest number.
Step 5: Otherwise, return b as the largest number.
Step 6: In the main program, take two numbers as input.
Step 7: Call the function and print the largest number.

Problem 05: Find the sum of numbers passed as parameters.

Step 1: Define a function named sum Numbers(a, b).
Step 2: Pass the numbers as parameters to the function.
Step 3: Add the numbers inside the function.
Step 4: Return the result of the addition.
Step 5: In the main program, call the function with the numbers.
Step 6: Print the returned sum.

Problem 06: Check whether a number is even or odd.

Step 1: Define a function named check_even_odd(n).
Step 2: Pass a number as a parameter to the function.
Step 3: Use the modulus operator % to check if the number is divisible by 2. Step 4: If n % 2 == 0, print that the number is even.
Step 5: Otherwise, print that the number is odd.
Step 6: In the main program, call the function with a sample input.

Problem 07: Perform basic arithmetic operations using functions.
Step 1: Define four separate functions: addition(a, b), subtraction(a, b), multiplication(a, b), and division(a, b).
Step 2: Pass two numbers as parameters to each function.
Step 3: In the addition() function, add the two numbers and return the result.
Step 4: In the subtraction() function, subtract the second number from the first and return the result.
Step 5: In the multiplication() function, multiply the two numbers and return the result.
Step 6: In the division() function, divide the first number by the second and return the result.
Step 7: In the main program, take two numbers as input from the user.
Step 8: Call each function and display the results of the operations.

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4. TEST RESULT / OUTPUT:

Problem01:

Print “Hello, World!” using a function.

Input:

```python
def hello_world():  1 usage
    print("Hello, World!")

hello_world()
```

Output:

```
uv run lab_04_p1_hello_world.py
Hello, World!
```

Problem02:

Multiply two numbers using a function:

Input:

```python
def multiply_numbers(a, b):  1 usage
    return a * b

result = multiply_numbers(2, 5)
print(result)
```

Output:

```
uv run lab_04_p2_multiply_two_num.py
10
```

Problem03:

Calculate volume and surface area of a cylinder using functions:

Input:

```python
import math

def volume(radius, height):  1 usage
    return math.pi * radius**2 * height

def surface_area(radius, height):  1 usage
    return 2 * math.pi * radius**2 + (2 * math.pi * radius * height)

r = float(input("Enter radius: "))
h = float(input("Enter height: "))

v = volume(r, h)
s = surface_area(r, h)

print(f"The volume is: {v:.2f}")
print(f"The surface area is: {s:.2f}")
```

Output:

```
uv run lab_04_p3_volume_surface_area_cylinder.py
Enter radius: 5
Enter height: 10
The volume is: 785.40
The surface area is: 471.24
```

Problem04:

Find the largest number between two numbers using a function:

Input:

```python
def largest_num(num1, num2):  1 usage
    if num1 > num2:
        print(f"Largest number is {num1}")
    else:
        print(f"Largest number is {num2}")


n1 = int(input("Enter first number: "))
n2 = int(input("Enter second number: "))

largest_num(n1, n2)
```

Output:

```
uv run lab_04_p4_largest_num.py
Enter first number: 4
Enter second number: 3
Largest number is 4
```

Problem05:

Find the sum of numbers passed as parameters:

Input:

```python
def sum_num(n1, n2):  1 usage
    return n1 + n2

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

result = sum_num(num1, num2)
print(f"Sum of {num1} and {num2} is : {result}")
```

Output:

```
uv run lab_04_p5_sum_num.py
Enter first number: 5
Enter second number: 10
Sum of 5 and 10 is : 15
```

Problem06:

Check whether a number is even or odd:

Input:

```python
def check_even_odd(n):  1 usage
    if n % 2 == 0:
        print(f"{n} is an Even number")
    else:
        print(f"{n} is an Odd number")


num = int(input("Enter a number: "))
check_even_odd(num)
```

Output:

```
uv run lab_04_p6_check_even_odd.py
Enter a number: 7
7 is an Odd number
```

Problem07:

Perform basic arithmetic operations using functions:

Input:

```python
def addition(num1, num2):  1 usage
    return num1 + num2


def subtraction(num1, num2):  1 usage
    return num1 - num2


def multiplication(num1, num2):  1 usage
    return num1 * num2


def division(num1, num2):  1 usage
    return num1 / num2


```

```python
n1 = int(input("Enter first number: "))
n2 = int(input("Enter second number: "))

result_add = addition(n1, n2)
result_sub = subtraction(n1, n2)
result_mul = multiplication(n1, n2)
result_div = division(n1, n2)

print(f"The result of addition of {n1} and {n2} is : {result_add}")
print(f"The result of subtraction {n1} and {n2} is : {result_sub}")
print(f"The result of multiplication of {n1} and {n2} is : {result_mul}")
print(f"The result of division {n1} and {n2} is : {result_div:.2f}")
```

Output:

```
uv run lab_04_p7_arithmetic_operations.py
Enter first number: 5
Enter second number: 5
The result of addition of 5 and 5 is : 10
The result of subtraction 5 and 5 is : 0
The result of multiplication of 5 and 5 is : 25
The result of division 5 and 5 is : 1.00
```

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5. Key Python Concepts Used

During the implementation of these programs, several important Python programming concepts related to functions and modular programming were applied.

Functions:
Functions were used to group a set of instructions into reusable blocks of code. Each task such as printing a message, multiplying numbers, or checking whether a number is even or odd was implemented inside a separate function.

Function Definition and Function Calls:
The keyword def was used to define functions, and the functions were executed by calling them in the main program. This demonstrates how programs can be organized into smaller manageable components.

Parameters and Arguments:
Some functions were designed to receive input values through parameters. These parameters allow functions to process different inputs without rewriting the code.

Return Values:
Certain functions used the return statement to send the result back to the main program. This helps store the computed result in variables and use it later in the program.

Modular Programming:
The cylinder problem demonstrated modular programming by dividing the program into two

separate functions: one for calculating volume and another for calculating surface area. This improves code organization and readability.

Conditional Statements:

Conditional statements (if–else) were used inside functions to make decisions, such as determining whether a number is even or odd or finding the largest number between two values.

Arithmetic Operators:

Arithmetic operators such as +, -, *, and / were used to perform mathematical operations inside functions.

User Input and Output:

The functions input() and print() were used to receive values from the user and display results clearly.

These concepts demonstrate how functions help organize programs efficiently and make code reusable and easier to maintain.

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6. Conclusion

This lab helped me understand how functions work in Python and how they can be used to make programs more organized. By solving these problems, I learned how to divide a program into smaller parts and reuse them when needed.

In the first problem, I created a simple function to print “Hello, World!”, which helped me understand how functions are defined and called.

In the second problem, I used a function to multiply two numbers, which showed how parameters and return values work.

The cylinder problem helped me understand modular programming by using separate functions for volume and surface area. This made the program easier to manage.

I also learned how to use functions to compare numbers, calculate sums, and check whether a number is even or odd. These problems helped me practice using conditions inside functions.

Finally, I used multiple functions to perform basic arithmetic operations, which showed how functions can be reused for different tasks.

Overall, this lab improved my understanding of functions and helped me write more structured and readable programs. It also gave me a good foundation for writing larger programs in the future.